

Requirements Traceability Matrix (RTM)

Legend:

- Requirement ID: Unique identifier for each requirement.
- Requirement Description: Description as per requirements.
- Business Priority: High, Medium, Low, or .
- Source: BRD/Stakeholder.
- Functional Area: Process/Module.
- Design Component: Logical/Physical module mapping.
- User Story/Feature Mapping: Backlog reference.
- Code Module/Repository Reference: Implementation location.
- Test Scenario ID: Reference to the test scenario.
- Test Case ID: Reference to the test case.
- Test Status: Not Started/In Progress/Passed/Failed/.
- Dependencies: Upstream/downstream dependencies.
- Risks: Associated risks.
- Comments: Notes or clarifications.

| Requirement ID | Requirement Description | Business Priority | Source | Functional Area | Design Component | User Story/Feature Mapping | Code Module/Repository Reference | Test Scenario ID | Test Case ID | Test Status | Dependencies | Risks | Comments |

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| REQ-001 | Read source csv data from volume and load to delta table customer and order | High | BRD | Data Ingestion | Delta Table Load | US-01 | modules/load_data.py | TS-101 | TC-101 | | Filesystem | Data file missing | |

| REQ-002 | Schema of customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId) | High | BRD | Schema Definition | Table Schema Config | US-02 | modules/schema.py | TS-102 | TC-102 | | REQ-001 | Schema drift risk | |

| REQ-003 | Add 'TotalAmount' column to order table (multiplication of PricePerUnit and Qty) | High | BRD | Data Transformation | Add Column Logic | US-03 | modules/transform.py | TS-103 | TC-103 | | REQ-002 | Transform error | |

| REQ-004 | Remove Null and Duplicate records from both tables | Medium | BRD | Data Cleansing | Clean Data Module | US-04 | modules/cleaning.py | TS-104 | TC-104 | | REQ-001 | Loss of valid data | |

| REQ-005 | Create 'ordersummary' table if not exists in catalog and schema | High | BRD | Table Management | Table Creation Script | US-05 | modules/tables.py | TS-105 | TC-105 | | REQ-004 | Duplicate table | |

| REQ-006 | Join customer and order data using CustId and load into SCD type 2 ordersummary table | High | BRD | Data Integration | SCD Type 2 Logic | US-06 | modules/scd2.py | TS-106 | TC-106 | | REQ-005 | Join error | |

| REQ-007 | Schema of 'ordersummary': CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date | High | BRD | Schema Definition | Table Schema Config | US-07 | modules/schema.py | TS-107 | TC-107 | | REQ-006 | Schema mismatch | |

| REQ-008 | Update SCD type 2 ordersummary table on customer change. Set old records Inactive, new Active, manage StartDate, EndDate | High | BRD | SCD Maintenance | SCD Type 2 Update | US-08 | modules/scd2_update.py | TS-108 | TC-108 | | REQ-006 | History loss risk | |

| REQ-009 | Create 'customeraggregatespend' table (Name, TotalAmount, Date) as per catalog and schema | Medium | BRD | Table Management | Table Creation Script | US-09 | modules/tables.py | TS-109 | TC-109 | | REQ-007 | Name collision | |

| REQ-010 | Aggregate 'TotalAmount' from 'ordersummary' by Name and Date, load into 'customeraggregatespend' | Medium | BRD | Data Aggregation | Aggregation Logic | US-10 | modules/aggregate.py | TS-110 | TC-110 | | REQ-009 | Incorrect totals | |

| REQ-011 | Implement steps 1–10 using delta live tables | High | BRD | Orchestration | Workflow Pipeline | US-11 | pipelines/dlt_pipeline.py | TS-111 | TC-111 | | ALL | Workflow failure | |

Definitions:

- US: User Story
- TS: Test Scenario
- TC: Test Case

All test status fields must be updated throughout the test lifecycle. All modules, files, and references are strictly based on requirements input where available; placeholders indicate areas pending further detail or input.